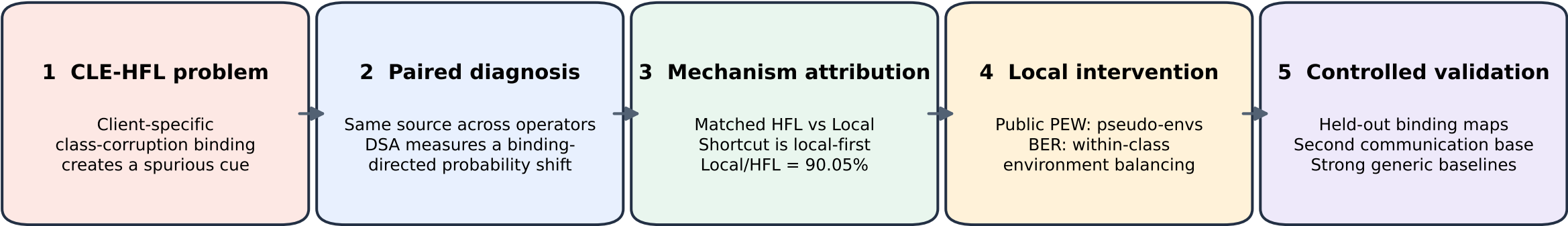
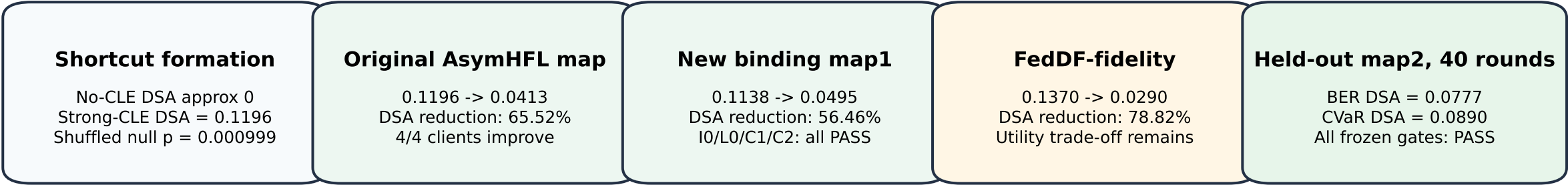


CLE-HFL: from directional shortcut identification to controlled mitigation

A problem-diagnosis-attribution-intervention-validation evidence chain



Formal evidence



Claim boundary

Supported: directional shortcut formation, local-first attribution, and mechanism-matched BER outperforming ERM, CVaR, and matched GroupDRO on held-out map2.
Not supported: taxonomy-free discovery, universal per-architecture dominance, or generalization across partitions, seeds, datasets, and real domains.